

Stage Grammar

6 modules and 36 lessons bundled as JSON. 4 interactive task types: Quiz, Spot-it, Sort-it, Build-it. Mistakes are never punished (explanations shown after 2 misses).

Constrained

mini 84

System prompt restricts Gemini to Socratic dialogue: probes reasoning and gives hints, never direct answers. Strict low token cap prevents lecturing; unlocks only post-task.

System Architecture

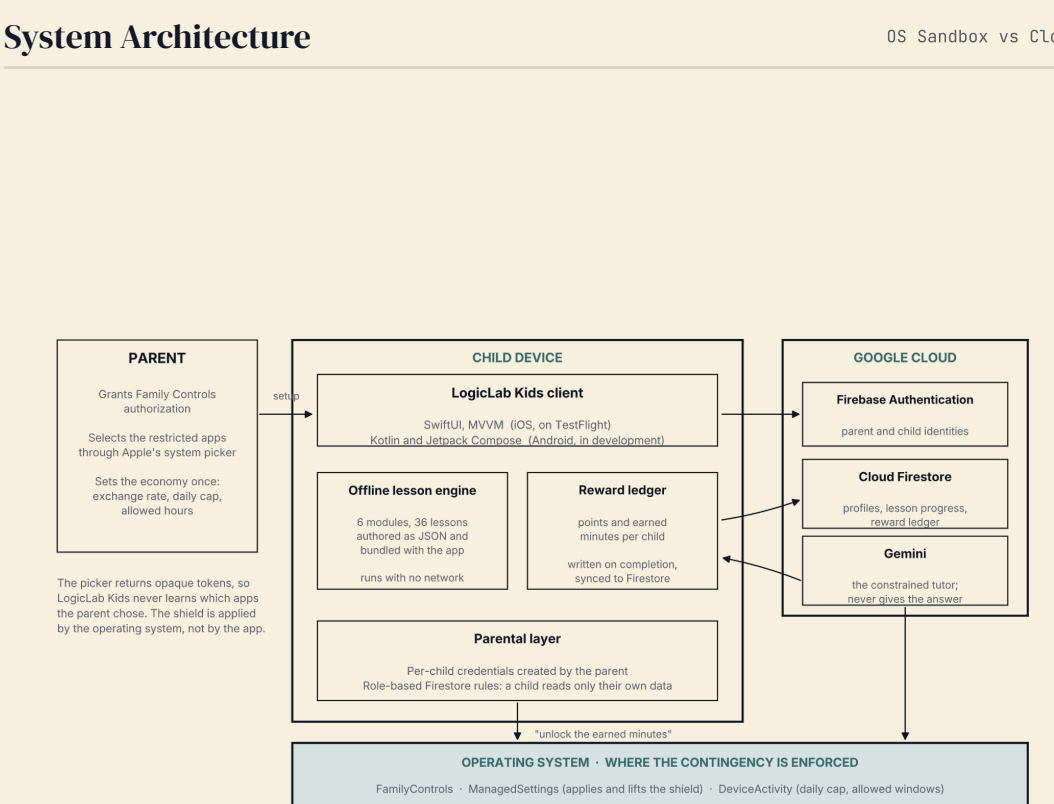


FIG 3. Hardware sandbox vs cloud: Screen-time contingency is enforced on-device via Apple FamilyControls and ManagedSettings. Cloud Firestore manages sync and parental rules.

Opaque Token Shielding

reasoning practice is among the lowest. The lock and key are in the OS: a child cannot negotiate around it.

Opaque Token Shielding
The system picker returns **opaque tokens**, so LogicLab Kids never accesses which apps they selected. Shields are applied directly by iOS Managed Settings, not in-app.

5-Stage Interaction Cycle

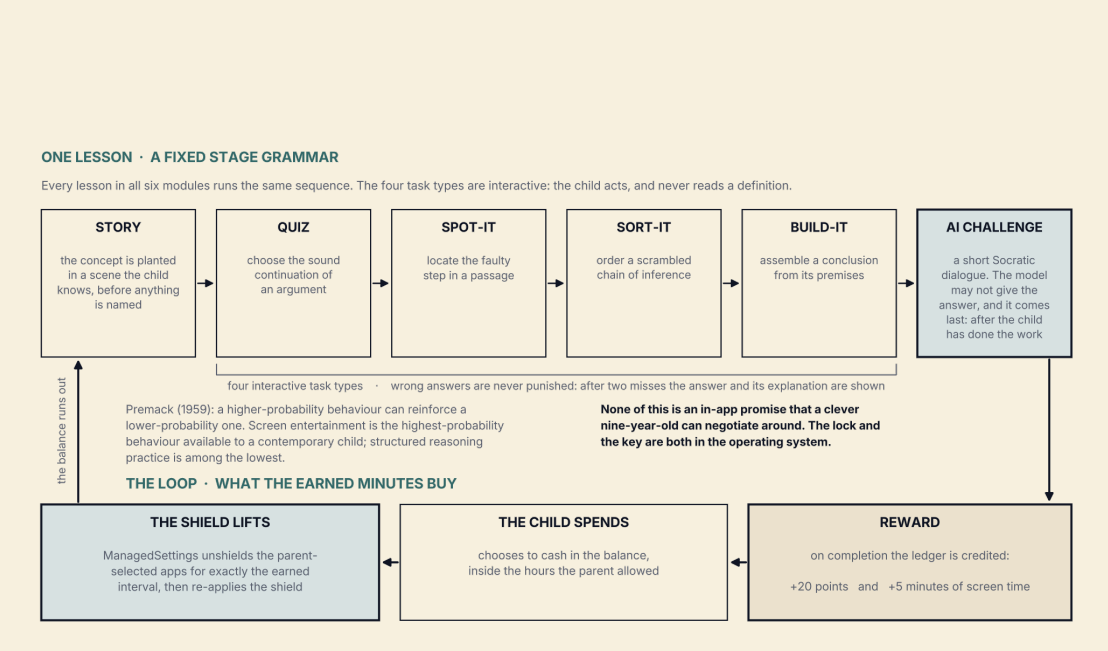


FIG 6. Fixed 5-stage lesson grammar (Story → Quiz/Spot/Sort/Walk → Socratic AI → Reward Credit → Shield Lift). Screen entertainment is unlocked strictly for the earned interval.

4 Core Task Types

assembly). The child acts dynamically rather than memorizing definitions.

+5 min
Earned Screen Time
+20 pts per active lesson

100%
OG-Level Shielding
Apple FamilyControls & Spectre

0 Net

Offline Runtime
Zero network required for lessons

1. Adobe Inc. 9